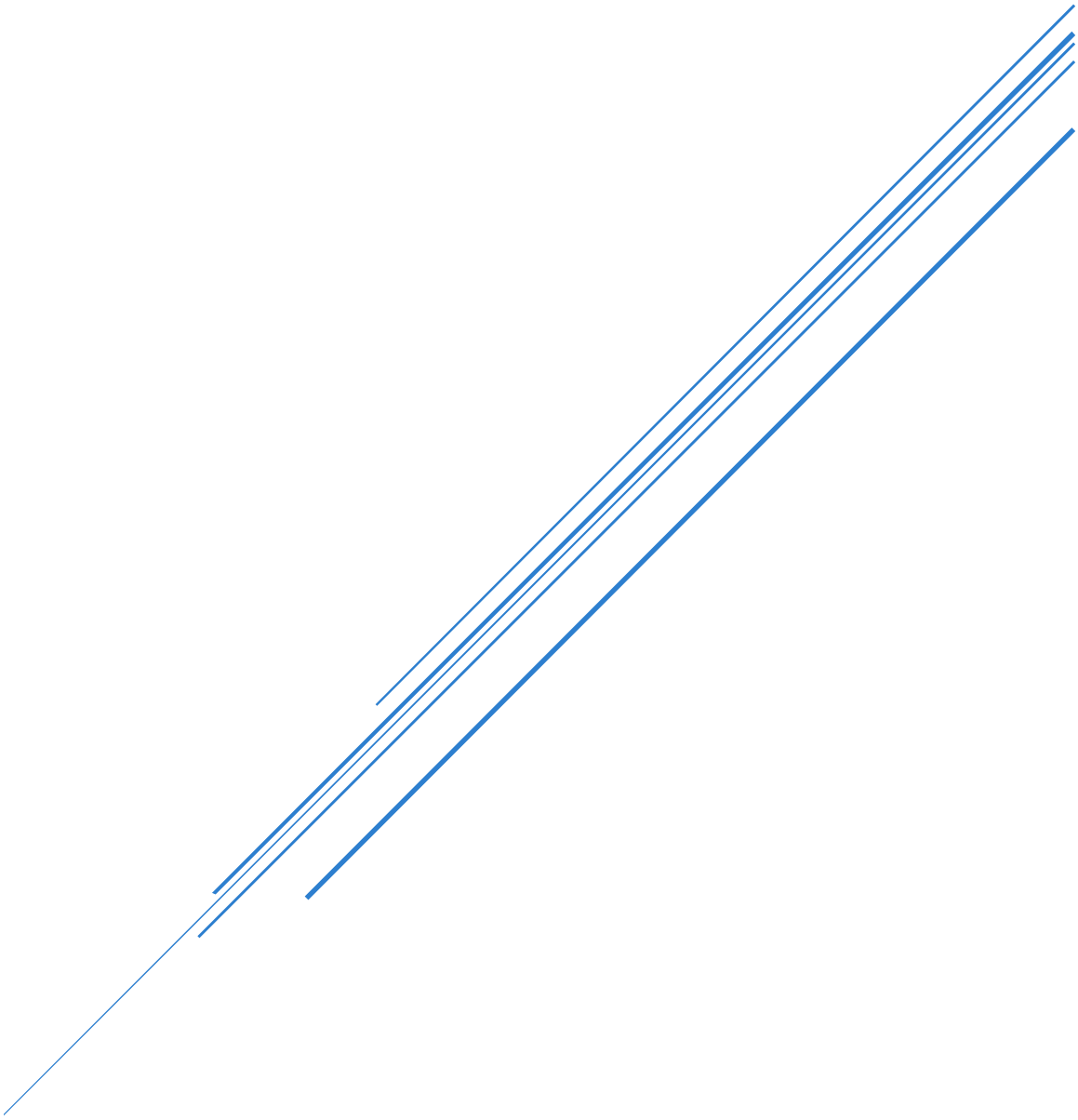
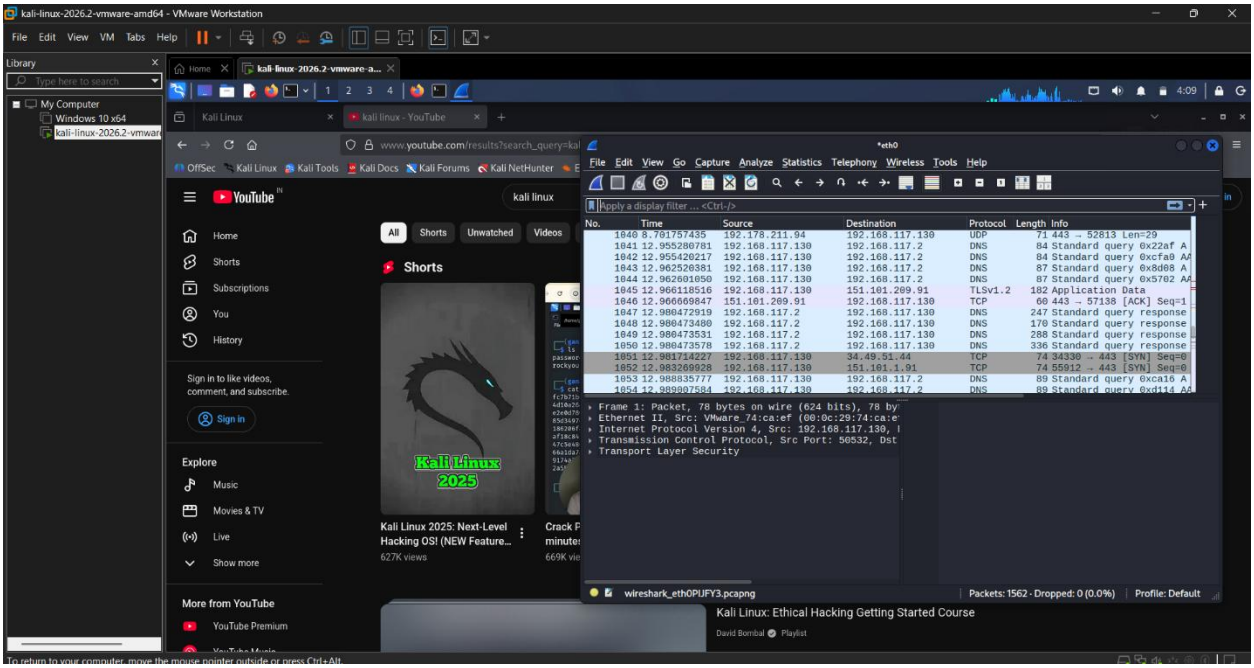
SOFTWARE AND HARDWARE TOOLS

Assignment - 7



1. **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

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I installed Wireshark on Kali Linux and captured network traffic for approximately 2 minutes while browsing YouTube. I selected the eth0 interface and saved the captured traffic.

Top 3 protocols observed:

1. **DNS – Resolves domain names into IP addresses.**
2. **TCP – Provides reliable data communication.**
3. **TLS – Encrypts communication between the browser and website.**

2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

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```
C:\Users\RAHUL>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [151.101.211.42] with 32 bytes of data:
Reply from 151.101.211.42: bytes=32 time=52ms TTL=60
Reply from 151.101.211.42: bytes=32 time=46ms TTL=60
Reply from 151.101.211.42: bytes=32 time=35ms TTL=60
Reply from 151.101.211.42: bytes=32 time=22ms TTL=60

Ping statistics for 151.101.211.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 52ms, Average = 38ms

C:\Users\RAHUL>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [23.55.244.186] with 32 bytes of data:
Reply from 23.55.244.186: bytes=32 time=16ms TTL=59
Reply from 23.55.244.186: bytes=32 time=18ms TTL=59
Reply from 23.55.244.186: bytes=32 time=16ms TTL=59
Reply from 23.55.244.186: bytes=32 time=23ms TTL=59

Ping statistics for 23.55.244.186:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 23ms, Average = 18ms
```

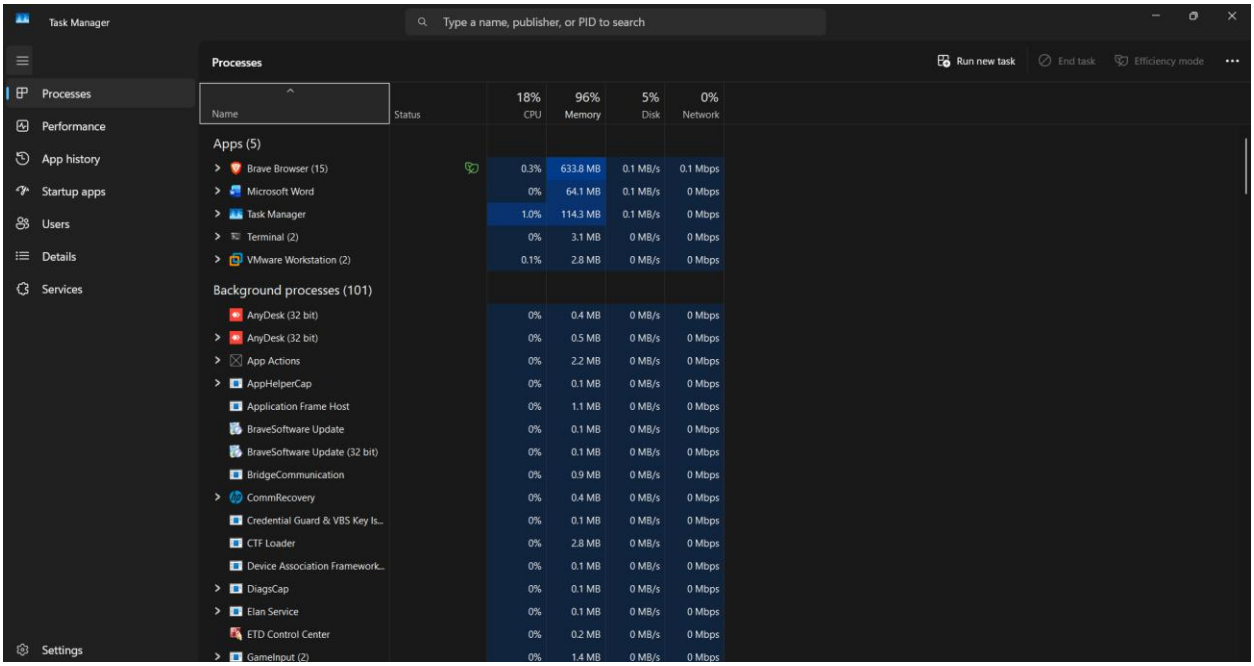
I used the ping command to test connectivity to Spotify and Zomato.

Website	Average Response Time
www.spotify.com	38 ms
www.zomato.com	18 ms

Result: Zomato responds faster than Spotify because its average response time is 18 ms, compared to Spotify's 38 ms.

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

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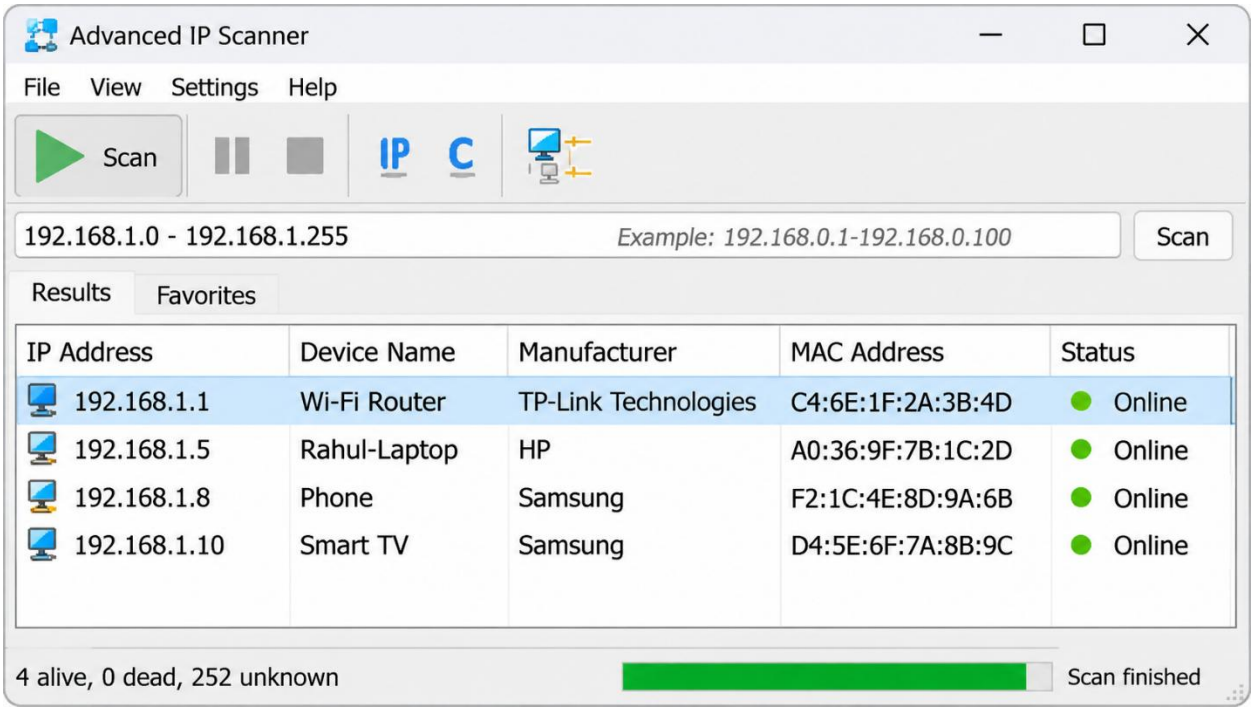


I opened Task Manager in Windows and checked the running processes and their memory usage.

Process Name	Memory Usage
Brave Browser (15)	633.8 MB
Microsoft Word	64.1 MB
Task Manager	114.3 MB
Terminal (2)	3.1 MB
VMware Workstation (2)	2.8 MB

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.**
   Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

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5. **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

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Hardware Tool	When to Use
1. Cable Tester	**Used to check network cables for faults, broken wires, or incorrect connections.**
2. Crimping Tool	**Used to attach RJ45 connectors to Ethernet cables when making or repairing network cables.**
3. Multimeter	**Used to measure voltage, resistance, and continuity while troubleshooting electrical issues in network devices.**